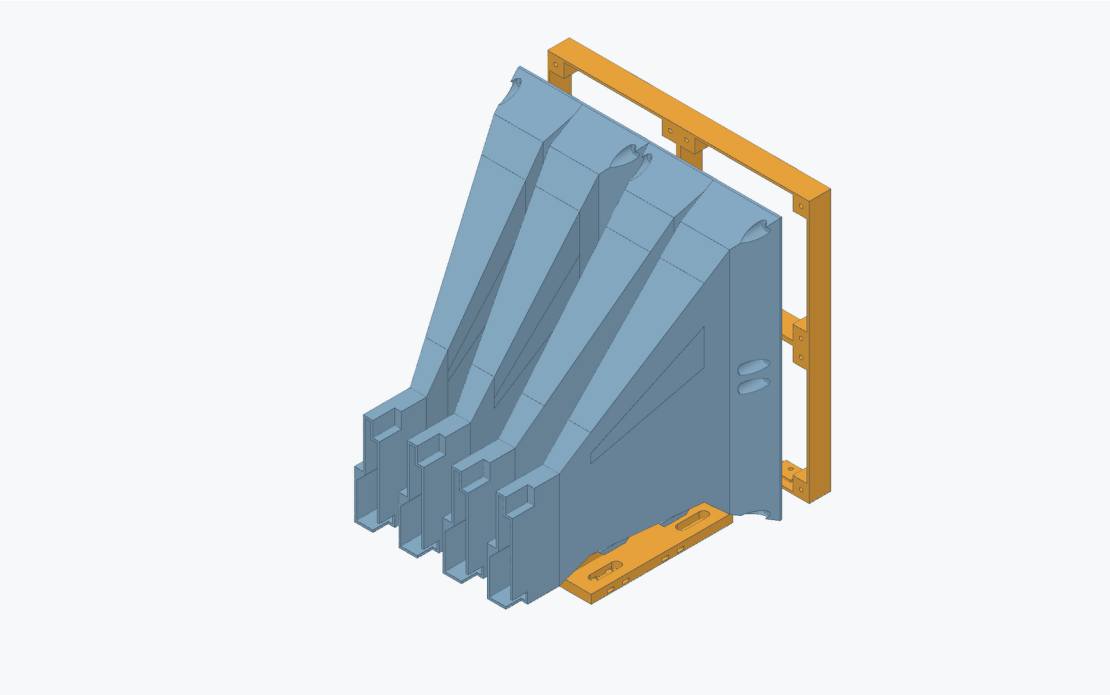
START HERE

Build the duct assembly

A bench guide for the parts in MOE_6U_Complete_STL_Print_Pack.zip.



Current CAD: two paired ducts with four GPU outlets. Gold identifies the supporting parts and fan-bank carrier. Fans are omitted; the gap at the carrier is occupied by the first fan stage.

Your build, as reported
Use four rods: front and rear crosswise rods plus both longitudinal rods. Leave the middle crosswise channel empty. Your rods are longer than the original 406.4 mm specification.

The rear PCIe bracket top flange has insufficient rearward clearance. This guide records that interference; it does not revise the printed parts.

Assembly route

1 Join and reinforce the chassis (p. 4). **2** Install the duct heatsets and assemble loose cradles (pp. 3, 5). **3** Bench-assemble the four fans, carrier and two ducts (p. 6). **4** Check card seating, lower in the duct unit and align it (p. 7). **5** Check clearances and airflow (p. 8).

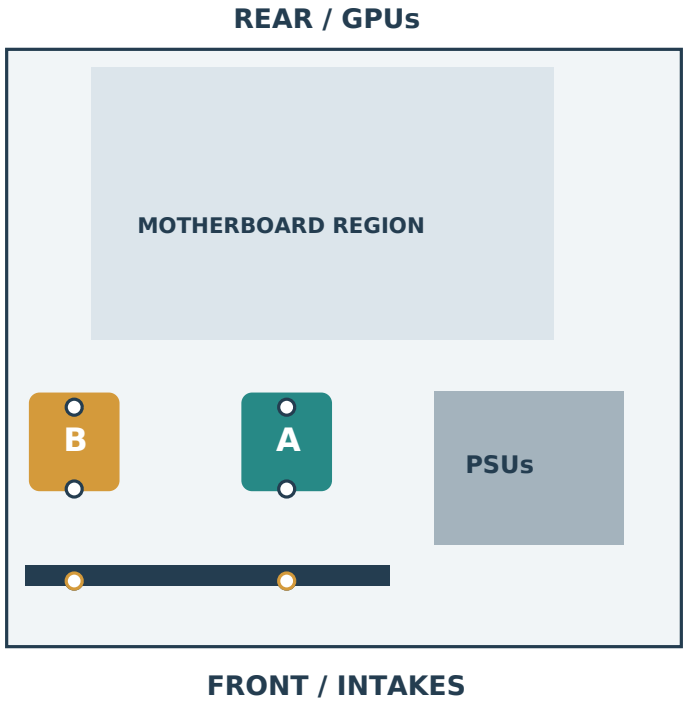
The ducts and fans carry their weight through the floor supports and carrier. The GPU outlets are interfaces, not structural hangers. No TPU or GPU attachment screws are used.

This is an assembly guide for the accepted geometry, with your later build notes added. Physical fit and the revised four-rod arrangement have not been load-tested.

IDENTIFY THE PARTS

Keep A/B and front/rear straight

All left/right directions below are as you stand at the chassis front, looking toward its rear.



PAIR B

Left fan column;
nearest left sidewall.

PAIR A

Right fan column;
nearest PSU side.

REAR support = GPU end

FRONT support = fan end

Plan is a location map, not a full component drawing. The six circles are the floor heatsets: four rearward anchors for the bases and the frontmost pair for the carrier.

STL name ending / group	Qty	Identification
PAIR_A_DUCT / PAIR_B_DUCT	2	Two tall, tapered paired bodies; keep A and B distinct.
PAIR_A_BASE / PAIR_B_BASE	2	Long rails. A is 64 mm wide; B is 70 mm wide.
...FRONT_SLIDER / ...REAR_SLIDER	4	Small blocks with two heatset pockets each.
...FRONT_RISER_10 / ...REAR_RISER_10	4	10 mm spacers; use these for the first assembly.
...FRONT_SADDLE / ...REAR_SADDLE	4	Wider top plates; one offset duct heatset each.
FAN_BANK_CARRIER	1	240 x 240 frame, 20 mm deep, two floor feet.
Optional_2mm_Height_Shim	-	Set aside initially; not a required extra layer.

Names above omit the Duct20_ prefix and _Print.stl suffix. Label the four saddles from their filenames before mixing parts: the offset holes differ. Compare any unidentified part with its STL in the slicer.

HARDWARE AND PREPARATION

Sort screws and fit heatsets

Quantities below cover the complete duct system with one stage of four 120 x 25 mm fans.

Connection	Hardware	Qty
Base rails to floor	M3 x 6 socket screws + washers	4 + 4
Sliders to base tracks	M3 x 10 socket screws + washers + nuts	8 + 8 + 8
Saddle / 10 mm riser / slider	M3 x 16 socket screws + washers	8 + 8
External duct ears to saddles	M3 x 8 socket screws + washers	4 + 4
Fan carrier to floor	M3 x 8 socket screws + washers	2 + 2
Four-fan stack	M3 x 55 bolts + washers + nuts	16 + 32 + 16
Heatsets, floor + moving parts	ruthex RX-M3Sx4.0, M3 thread, 4 mm long	6 + 12

Duct total: 42 M3 screws/bolts, 58 washers, 24 nuts and 18 heatsets, including the six floor inserts. Already installed inserts count toward that total. Chassis splice hardware is separate (p. 4).

Heatset locations

Part	Inserts	Where
Front-left floor	6	Two columns of three blind holes; install flush.
Four sliders	8	Two blind round pockets in the top of each slider.
Four saddles	4	One offset blind pocket in the top of each saddle.

1. Clean and identify. Remove strings and loose bridge debris. Clear passages and through-holes. Sort the M3 lengths and keep each A/B front/rear saddle labeled.

2. Install inserts before electronics. Support each piece flat. Heat the insert straight into its pocket until flush with the surface; let it cool before threading a screw. Keep molten plastic out of the thread. The bases, risers and duct bodies do not receive these inserts.

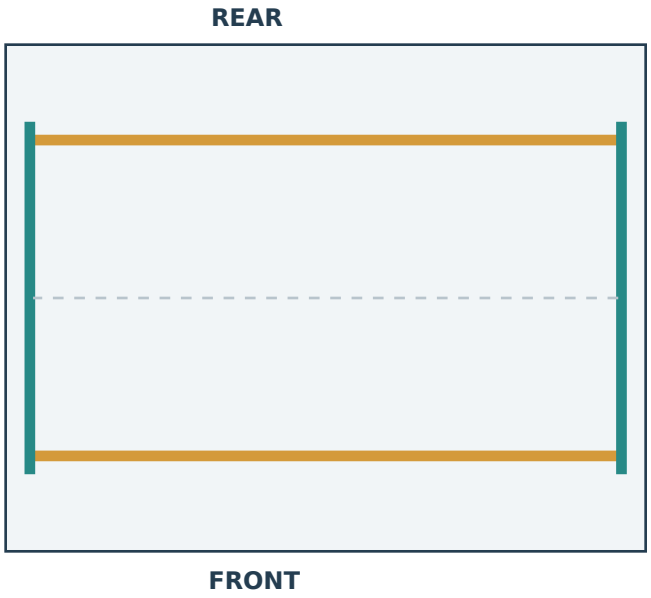
3. Load the track nuts. Slide two M3 nuts into each base station, one in each side-loading channel. Eight nuts total. Position them beneath the transverse clamp slots before the bases are installed.

> **Hardware fit**
> The model assumes washers 0.5 mm thick and no more than 7 mm OD; M3 nuts 5.5 mm across flats and about 2.4 mm thick; socket heads no more than 5.5 mm diameter for the ducts. Use a 5.5 mm nut driver with a slim nose, ideally no more than 8 mm OD, at the fan corners. Short coarse-thread fan screws do not fit the duct mounting scheme.

CHASSIS FIRST

Your four-rod configuration

Complete the underside hardware before fitting the motherboard and GPU assembly.



2 CROSSWISE RODS

Front + rear locations

2 LONGITUDINAL RODS

One along each side

MIDDLE CHANNEL EMPTY

Your as-built arrangement

Diagram shows channel locations only; actual rod ends depend on your longer stock. All four rods remain at the same base height.

1. Join the quadrants. On a flat surface, lower rear-left into rear-right, then front-right onto rear-right, then front-left into both adjacent joints. Align the vertical dovetails/tongues without spreading the walls.

2. Fit the seven connectors. Install four underside bridges with 8 M3 x 14 screws, 8 nuts and 16 washers. Install the left, right and rear upper straps with 6 M3 x 12 screws, 6 nuts and 12 washers. Keep base nuts, washers and screw tips recessed below the interior floor.

3. Insert only front and rear crosswise rods. Feed from the left side toward their blind stops. Leave the middle crosswise channel vacant. Then insert both longitudinal rods from the front. Use a blunt pusher and hand pressure, not a hammer.

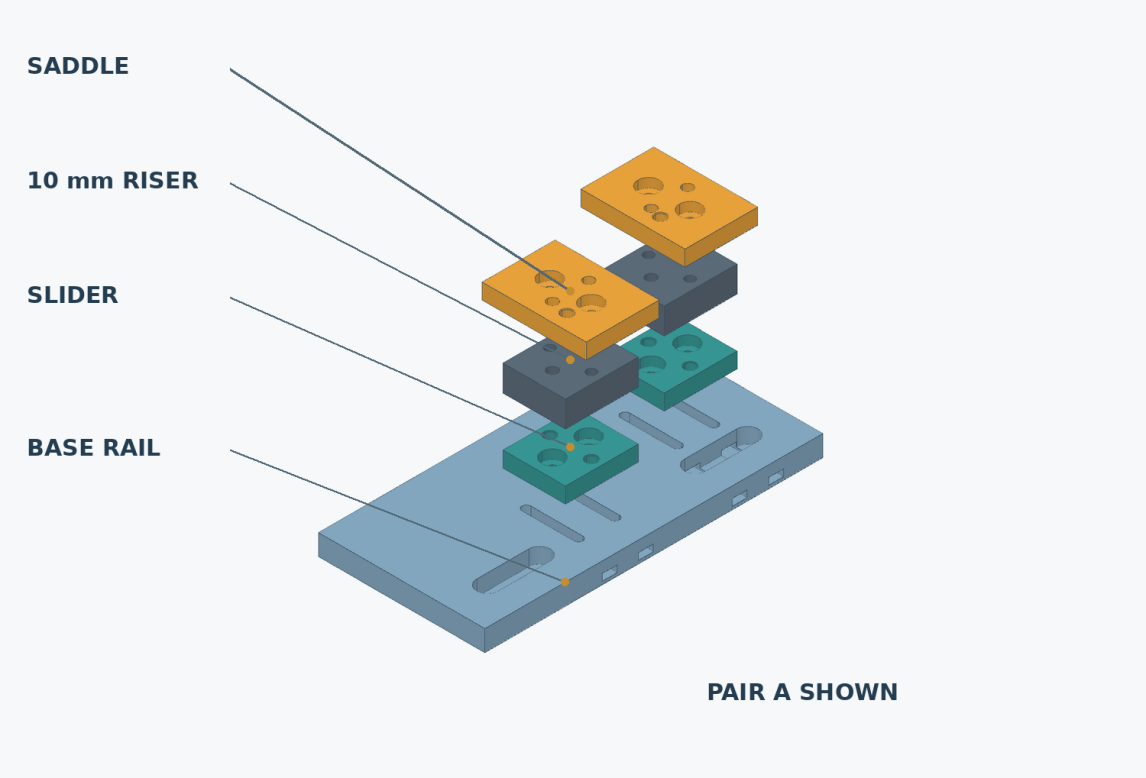
4. Check the ends and retain them. The original two longitudinal stop positions use 2 M3 x 18 screws, 2 nuts and 4 washers. With your longer rods, confirm both stop holes remain clear and the bolts fit without bearing on a rod. If a rod blocks a stop bolt, leave that fastener out pending a fit check; do not force it through the steel or plastic. Retention is incomplete until that conflict is resolved.

> **Before loading the chassis**
> The omitted middle rod means this is not the original five-rod reinforcement arrangement. Use full base support on the bench and rack shelf. Check that the seams stay closed, all rods are contained, and the front wall seam is bridged by the installed middle front fans. Do not lift the loaded assembly by a wall or duct.

BUILD THE SUPPORTS

Assemble four cradle stacks

Each pair has one base rail and two stacks: rear toward the GPUs, front toward the fans.



Actual Pair A CAD, exploded vertically. Pair B uses the same layer order. Keep all counterbores and insert openings facing upward.

1. Fit the sliders to the bases. At each station, use two M3 x 10 screws and washers in the slider counterbores, down through the transverse base slots into the track nuts. Leave them just loose enough to slide sideways.

2. Add the 10 mm risers and saddles. Use two M3 x 16 screws with washers per stack. They pass down through the saddle counterbores and riser into the two slider heatsets. Snug these stacks flat; the lower clamp screws provide lateral adjustment. The small clearance holes in the riser/saddle preserve access to those lower clamps.

3. Orient each saddle. Its single offset heatset goes on the front/fan side of the stack. Keep the correct A/B and front/rear saddle at its named station. The two recessed riser screws are separate from the single duct-mount insert.

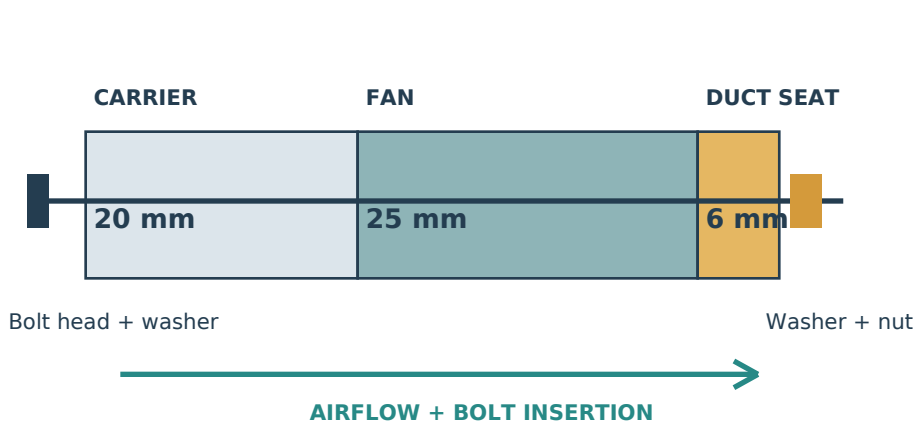
4. Mount the two bases loosely. Use two M3 x 6 screws and washers per base into the four rearward floor heatsets. Leave the frontmost two floor inserts for the carrier. Start with the floor slots near their middle and use the assembled duct unit to set the final positions.

Coordinate reference if needed: A floor anchors X260; B floor anchors X400. Each base uses Y350 and Y430; the carrier uses Y520. X is measured from the right exterior and Y from the rear exterior. These are identification datums, not new holes to drill.

BENCH ASSEMBLY

Fans, carrier and ducts

Start with four ARCTIC P12 Pro A-RGB fans in a 2 x 2 bank. Air moves from the chassis front into the GPU ducts.



One-stage stack, shown front to rear. The carrier remains in the stack even when the optional second fan stage is absent. The fan bank is separate from the six front-wall intake fans.

1. Arrange the two ducts. Pair B is the left column, Pair A the right, as seen from the chassis front. Support their bodies on the bench without loading the thin GPU noses. Place two fans at each large duct face.

2. Set airflow and cable exits. Orient all four fans to blow through the ducts toward the GPUs. Use the fan frame airflow arrows where present. Route the leads toward accessible outside edges before adding the carrier; do not trap a wire between frames.

3. Add the common carrier. Place the 20 mm carrier against the fans' intake faces, with its two floor feet down. Align all sixteen mounting holes. It ties the two duct columns into one bank.

4. Install the long bolts. Use one M3 x 55 bolt at each corner, with a washer under its head. Insert from the front through carrier, fan and the 6 mm duct seat. Add a washer and M3 nut in the duct's rounded corner recess. Hold the nut with a slim 5.5 mm driver while turning the bolt from the front.

5. Tighten evenly. Start every nut before snugging the corners in a diagonal pattern. Close frame gaps without bowing the printed face or fan frame. Confirm each nut is fully engaged and no screw bottoms out or projects into moving parts. Turn every rotor by hand to check for rubbing.

> **Do this on the bench**
> The corner recesses are most accessible before installation. The 2 mm divider at each duct fan face must clear the actual rotor. A mounted fan that rubs is not ready to power.

INSTALL AND ALIGN

Let the supports carry the load

The rigid ducts fix the GPU outlets relative to the fan bank. Cradle adjustment seats the supports under that geometry.

1. Establish card position first. Fit the motherboard and GPUs before final duct alignment. Check the lower bracket tabs in their rear channel and full, level PCIe connector engagement. Because the top flange binds rearward in your print, inspect that area closely while the ducts are still out.

> **Known PCIe flange interference**
> Do not use a rear retention screw or the duct to pull a card into the slot, bend the bracket into line or overcome the obstruction. If the flange prevents full seating or secure retention, pause that fitment and document where it touches. This guide makes no chassis or bracket modification; the eight-position rear opening is only an aiming reference.

2. Lower in the fan-and-duct unit. With the four cradles loosely located, bring the assembled unit into the chassis from above. Support the weight by the fan frames and substantial duct walls. Watch all four GPU outlets, power plugs and cables; do not slide the noses into the cards under load.

3. Start the carrier screws. Use the two M3 x 8 screws with washers through the carrier feet into the frontmost floor heatsets. Start both threads loosely so the carrier is located without forcing the bank sideways.

4. Bring the cradles into contact. Slide the base rails fore/aft and the lower sliders sideways until all four saddles sit beneath the external duct feet. Use the four nominal 10 mm risers. Both feet of each paired duct should rest evenly, with no rocking or preload against a GPU.

5. Attach the ducts to the saddles. Install four M3 x 8 screws and washers from above, through the external diamond holes between the two passages of each duct, into the saddle heatsets. Two screws per paired duct. No screwdriver passes through an air passage; no screw goes into a GPU.

6. Check, then snug. With all screws started, check the four outlets for even seating/clearance and verify that no card moves when the supports are tightened. Snug carrier feet, base-floor screws, slider clamps, riser screws and finally the four duct-ear screws. Recheck alignment after each group.

If a stack needs height adjustment: the optional 2 mm shim is for a deliberate replacement riser stack, not an automatic addition to the 10 mm riser. Riser screw length follows total riser thickness + 6 mm. Raising a support must not lift the fixed fan bank or press the GPU noses into the cards. A mismatch that requires bending the rigid duct cannot be solved with these clamps.

FINAL FIT AND FIRST RUN

Check once before power

A short final pass now is easier than pulling the fan bank back out later.

[ ] Structure. All seven chassis connectors are fitted; the middle rod channel is empty; the remaining rods are contained and their retention is confirmed. The chassis sits on full base support.

[ ] Cards. Each card is fully seated and independently retained. The reported rear-flange interference is not being held under screw tension. Duct weight is carried by the supports and carrier.

[ ] Ducts. Both feet of each duct sit on their saddles. All four external screws are snug, inserts remain flush, and no body is being bent to meet a hole.

[ ] Fans. All four duct rotors turn freely. All four point toward the GPUs. The six front fans are intake; the rear CPU fan is exhaust. Frame screws and the duct-face divider clear the rotors.

[ ] Wiring. No lead is pinched at a frame or under a foot. Secure loose leads outside the airways and fan sweep. Connect PWM/ARGB through the intended controller and observe its port power limits.

[ ] Clearances. Check GPU power plugs, the top rim/lid space and the front intake fans. The nominal 240 mm bank leaves only 1.3 mm below the chassis rim; actual print and fan dimensions govern.

First run

With the fit checks complete, start the fans at a low commanded speed and listen for rubbing. Confirm each GPU passage has flow in the intended direction. Increase fan speed, then add GPU load gradually while watching every card's reported temperature. Stop for a stalled fan, rubbing, a shifting duct or loss of card seating. Thermal performance still needs this physical test.

Optional second stage - later

Add four more 120 x 25 mm fans in front of the carrier, pointing the same way as the first stage. Replace the sixteen M3 x 55 bolts with M3 x 80 bolts; keep one washer under each head and one under each corner nut. The stack becomes **head/washer - extra fan - carrier - first fan - duct seat - washer/nut**.

The CAD allowance is 4.2 mm between the extra fan frames and the existing front intake fan reference; with the specified hardware, bolt-head clearance falls to about 0.7 mm. Verify that gap physically before installing or powering a second stage.

For service

Disconnect the fan leads, remove the two carrier-floor screws and four duct-ear screws, then lift the fan-and-duct unit while supporting its weight. Leave the bases and cradles in place if their alignment is good. Reach the corner nuts on the bench.

Basis: Rev24 dimension document; Step20 duct parts/source and assembly notes; your 4 October four-rod and PCIe flange reports. Earlier five-rod renders and Step17/18 support/TPU instructions do not describe this as-built assembly.